Response to Reviewer 2

Manuscript: *Proxy-Based Diagnostics of Quantum Oracle Sketching Robustness for Non-IID Sensor and Telemetry Streams*
Journal: *Sensors* (MDPI) · **Manuscript ID:** sensors-4470240

Dear Editor,

Thank you for the opportunity to submit a revised version of our manuscript, *Proxy-Based Diagnostics of Quantum Oracle Sketching Robustness for Non-IID Sensor and Telemetry Streams* (Manuscript ID: sensors-4470240), to *Sensors*. We appreciate the time and effort the reviewer has dedicated to the paper, and we are grateful for the constructive comments, which have improved it considerably.

We agree with the central point of the review. The previous version, although hedged internally, still read as evidence about quantum performance, whereas what the work can honestly support is a proxy-based diagnostic and hypothesis-generation framework for correlation-sensitive streaming analysis. The revision repositions the manuscript accordingly, in the title, abstract, statistics, figures and conclusions, without removing any experiment or altering any measured result. All stochastic outputs regenerate identically from the same seeded pipeline, which we verified before beginning the re-analysis. In accordance with the Academic Editor's request that preprints not be cited, the Zhao et al. (2026) preprint has been removed from the manuscript and the reference list, which now contains 54 references; no other work was substituted for it, and the affected statements were revised so that each remaining reference supports only what it actually establishes.

The changes are marked in the highlighted version submitted with this letter. A point-by-point response to the comments follows.

Comments from Reviewer 2

Comment 1: *The key issue is the relationship between τ and r and the quantities in the framework of quantum oracle-sketching. ... They have to strengthen the theoretical basis for the chosen proxy construction and distinguish the results from the original from the assumptions that are used in this paper.*

Response: Thank you for identifying this as the key issue. Section 3.3 now addresses it directly and states the boundary rather than assuming an equivalence. The section has five labelled parts. The first sets out the published quantum-streaming results that motivate the study, namely a polynomial quantum space advantage for one-pass triangle counting, an exponential quantum space advantage for approximating maximum directed cut, and a black-box route from certain classical sketches to quantum streaming algorithms, together with the statement that these are problem-specific results that do not establish the operational tau, r, T_eff or accuracy model used here and that this paper adds no quantum theorem. The remaining parts cover the assumptions this paper adopts, the heuristic constructions it introduces, the memory curves retained as illustrative scaling references, and the scope limitations, which state that tau and r are measurement conventions and are not formally equivalent to a universal mixing time, an effective-sample count or a quantum sample-overhead parameter. Table 1 gives the provenance of each quantity with an explicit not-guaranteed column. In direct answer to your question, tau and r are operational summaries defined in this paper for the proxy-based diagnostic; the manuscript neither implements the framework nor claims to prove or reproduce any quantum result.

Comment 2: *The terminology of quantum performance should be moderated throughout the paper. ... we will refer to them as proxy-based predictions instead of the actual quantum performance. This is particularly* *relevant in the Abstract, Results, Discussion, figure captions and Conclusion.*

Response: Agree, and a systematic terminology pass was made over each of the surfaces you list. The term "quantum-favourable" no longer appears and has been replaced by "proxy-favourable" or "hypothesized favourable". Constructions such as "quantum performance proxy crosses below classical performance" have been rewritten, for example as "the accuracy proxy computed with the analytic tau lies below the measured classical baseline". The relevant column of Table 6 is now headed "Proxy hypothesis". Because captions alone cannot correct text that is part of an image, Figures 1, 2 and 6 were regenerated to remove in-image labels referring to qubit counts and quantum order-notation. Statements that no quantum algorithm is implemented now appear in the Abstract, in Section 3.3, in four results subsections, in the Threats section and in the Conclusions.

Comment 3: *The statistical comparison between classical baselines and the proxy model is still not resolved. Our Wilcoxon tests compare stochastic classical results for seeds to a proxy value that is deterministic for a given value of parameters. ... We believe the magnitude of the classical minus proxy difference and the uncertainty may be a better gauge than simply p-values.*

Response: We agree, and have adopted this in full, so that magnitude and uncertainty now lead and the tests are re-scoped to what they can support. For each sweep point, Table 4 reports the Hodges-Lehmann estimate of the paired difference with its exact signed-rank 95% interval, for example +0.042 with interval [+0.011, +0.087] at rho = 0 and -0.046 with interval [-0.079, -0.003] at rho = 0.88, together with the rank-biserial correlation, which we interpret as sign concordance across seeds because the comparison is against a near-constant reference and all magnitude information resides in the difference estimate. The p-values are retained only as descriptive companions. We disclose that none survives Holm correction at n = 10 and why, and we disclose the sensitivity of the integer-valued tau estimator, where one lag step corresponds to 0.012 in accuracy and is sufficient to move the interval at rho = 0.48 across zero. The interpretation paragraph states that the crossover is the deterministic solution of the equation between the proxy accuracy and the mean classical accuracy, with uncertainty inherited from the estimator and the constants rather than from the tests. The full re-analysis, which first reproduced the published per-seed results to six significant digits, is included with the revision.

Comment 4: *The huge discrepancy between the analytic and empirical refreshing time is one of the most important findings in the paper, especially for the long-range dependence regime. ... The authors should discuss this in more detail and discuss its implications for the interpretation of the central (τ,r) sensitivity landscape.*

Response: We agree that this is one of the paper's more important findings and have promoted it accordingly. The new Section 7.3 gives the mechanism: the analytic proxies target model-level worst-case scales, that is joint-chain mixing and tail-dominated integrated autocorrelation, whereas the empirical estimator measures the short-lag marginal decorrelation of the binarised stream, and for a non-mixing process there is no scalar on which the two could agree. The consequences for the landscape are carried through the paper. Verdicts are now bound to their estimator convention wherever they appear, and we state openly that the burst and long-memory extremes exchange places between the analytic and empirical conventions, because burst saturates through its measured duplication while the high empirical value for long memory is a short-lag artefact. This appears in Experiment 9, in Section 7.3, in the answers to RQ2 and RQ3, and in the captions of Figures 3 and 7. The abstract now says that the two estimates diverge, rather than that one overestimates the other, and adds that they answer different questions.

Comment 5: *The classical-memory comparison needs to be taken with care. The classical lower bound curve ... is not actually an empirical lower bound for the SGD, averaged-SGD, or Count-Min tools used in our experiments. ... we should make it more prominent in the dimension-scaling discussion.*

Response: Agree, and this is now made prominent in exactly that discussion. The curves are no longer presented as cited bounds at all. Experiment 6 opens with the statement that no memory lower bound is measured empirically in the experiment, that the curves in panel (a) are illustrative quantities, and that no result constitutes an empirical memory floor for online SGD, averaged SGD or Count-Min. The caption of Figure 6 labels panel (a) as an illustrative memory-scaling reference that is paper-defined and contains no measured content, and states that the exponential divergence of the curves is a visual scaling contrast rather than an established separation. Section 3.3 records under "Memory references" that the published quantum-streaming papers do not establish this pair of curves for the diagnostic task studied here and that they are retained solely as a paper-defined illustrative reference. Table 1 carries the same status, and Section 7.1 states that none of its structural factors is a measured quantum result.

Comment 6: *The real-stream analysis is useful as a sanity check but it is limited to two NAB datasets. The conclusions on the sensor and telemetry applications should therefore be conservative.*

Response: We agree, and the conclusions drawn from the real streams are now explicitly conservative. Section 6.8 states that the experiment is illustrative rather than statistically representative, the external-validity paragraph records that generalisation is unverified, and the abstract claims placement only. In addition, a new encoder and resolution sensitivity table strengthens what can legitimately be said from two streams: across ten re-encodings of the same raw series, using three sampling resolutions and three bucket widths, the ordering of NYC Taxi above Machine Temperature holds in every configuration, so the reported placement is at least stable with respect to the encoder. Broader corpora are committed to in Future Work, with a concrete test protocol given as hypothesis H1.

Comment 7: *the target defined based on a percentile of the full time series should also be reconsidered in the context of streaming analysis, since using the full series to set the target threshold can take into account future observations. A threshold set from a first calibration interval or updated only from previously observed samples would provide a more rigorous prequential design.*

Response: Thank you for this observation, which we consider well taken, and which we now disclose explicitly rather than leave implicit. Section 6.8 states that the threshold is computed from the full series and therefore uses information that would not be available prequentially at each step. It also states that the feature stream and the tau and r estimates involve no such look-ahead, so the landscape placement, which is the central output for these streams, is unaffected, and the leakage touches only the classical baseline reference numbers. We further state that a threshold calibrated on an initial interval, or updated only from previously observed samples, is the more rigorous prequential design, and we adopt it as future work. The Threats section records this as a residual risk under internal validity. We chose disclosure over re-running the baselines in this revision in order to keep every published number reproducible, and we would be glad to re-run them with a calibration-interval threshold in a further round if you prefer.

Comment 8: *The interpretation of the NYC Taxi Count-Min result also needs to be qualified. The six-bit representation gives a relatively small number of possible feature vectors and hence memorising the same feature patterns is relatively easy. ... I would like to give more weight to the implications when we think about the apparent agreement between the Count-Min performance and our proxy prediction.*

Response: Agree, and the qualification now carries weight in three places. The dedicated paragraph explaining why Count-Min performs well is retained, and the interpretation now states plainly that the favourable-side separation between the two streams rests on Count-Min alone and that the hashed linear learners do not corroborate it. We additionally report that both SGD variants fall below the majority baseline on accuracy and that the balanced accuracy of averaged SGD straddles chance, so the apparent agreement between Count-Min and the proxy prediction is explicitly not read as validation. Section 7.4 lists representation-specific effects of this kind among the phenomena the landscape cannot anticipate, and, going further in the direction you indicate, the same section concedes that, read strictly as an accuracy projection, the proxy is falsified on the Machine Temperature stream, where 0.500 is predicted and 0.72 to 0.80 observed, with the rare-class diagnostic reading offered as interpretation rather than confirmation. Table 6 now carries the ordering-to-ordering intervals, with Count-Min F1 at 0.644 plus or minus 0.062 and a per-ordering range of 0.43 to 0.72, which tempers the headline number.

Comment 9: *In particular, the target function ablation should be highlighted because it shows that our classical performance can change dramatically with the classification task in question and that our proxy is in fact target independent. A correlation-based proxy that is insensitive to the target function cannot necessarily predict the performance of algorithms for different kinds of decision problems.*

Response: We agree, and this is now highlighted and stated as a scope boundary in your own terms. Section 6.9 states that classical learner performance is strongly target-dependent while the proxy depends only on tau, r and T, and is target-independent by construction, so that the proxy cannot universally predict downstream classifier performance and instead characterises the correlation structure of the stream, that is the effective sample budget available to any bounded-memory learner, rather than task-specific learnability. Section 7.4 elevates this to the list of things the landscape cannot do, adding the empirical observation that measured classical accuracy is flat while T_eff falls fivefold, so the budget model does not predict even the classical arm's accuracy. The caption of Figure 9 states that the proxy does not track task-specific learnability, and the answer to RQ2 now describes tau and r as budget summaries rather than accuracy predictors. The protocol of the ablation, which is prequential with a learning rate of 0.01 and T = 4096, is also stated so that its absolute levels are not misread against the main experiments.

Comment 10: *the strongest contribution of the paper is the (τ,r) sensitivity landscape proposed as a hypothesis-generating tool ... The paper would be better off having this methodological contribution as the main message rather than the separation of quantum-classical performance.*

Response: Thank you for this closing recommendation, which we have taken as the identity of the revised paper. The title leads with "Proxy-Based Diagnostics"; the abstract announces a proxy-based diagnostic and hypothesis-generation framework; the contribution list names the landscape as the paper's central diagnostic and hypothesis-generation artifact; Section 7.4 states the falsifiable hypotheses it generates; and the Conclusions open by identifying the landscape as the primary contribution and hypothesis generation as the secondary one, followed by the sentence that the paper demonstrates no quantum advantage and implements no quantum algorithm.

We hope that the revised manuscript now meets the standards of *Sensors*, and we look forward to hearing from you in due course. We would be glad to respond to any further questions or comments.

Sincerely,

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